

Edmonton Urban Canopy Cover Analysis 2025

Project Report

City of Edmonton | Parks & Roads Services | Urban Forest Initiatives

Date: March 2026 **Data Source:** 2025 City of Edmonton Vegetation LiDAR **Program Lead:** Erin Belva, Senior Program Lead **Technical Lead:** Travis Kennedy, P.Ag

1. Executive Summary

A comprehensive urban canopy assessment was completed using the City of Edmonton’s 2025 vegetation LiDAR dataset (3,233 tiles, 480 GB, 25 pts/m2, leaf-on collection). The analysis produced city-wide canopy cover metrics, individual tree delineation, conifer/deciduous classification, top-5 species identification, and BurnP3-ready FBP fuel type mapping.

Key Findings

Metric	Value
City-wide canopy cover	15.2% (119.6 km2 / 784.5 km2)
Individual trees and shrubs detected	5,168,142
Conifer/deciduous classification accuracy	90.4%
Top-5 species classification (macro F1)	0.74
Elm genus detection (F1)	0.82
Total fuel area mapped (FBP)	35,768 ha

Edmonton’s canopy cover has increased from the 2009 UFORE baseline of 10.3% to 15.2%, representing significant progress toward the UFMP target of 20%.

2. Data Sources

Dataset	Source	Details
Vegetation LiDAR	2025 collection	3,233 tiles, 25 pts/m2, EPSG:3776
Orthophoto	Edmonton Open Data	7.5 cm GSD RGB, 199 tiles
Tree Inventory	Edmonton Open Data	474,289 city-managed trees with species
UPLVI	Edmonton Open Data	12,825 ecological classification polygons
Naturalized Areas	Edmonton Open Data	3,108 naturalized vegetation polygons
Elevation (CDEM)	NRCan	30 m DEM, NTS sheet 083H

3. Methodology

3.1 Canopy Cover Analysis

Height-normalized vegetation rasterization at 1 m resolution following USFS i-Tree / American Forests methodology: - Ground DEM built from LAS class 2 (ground) points using binned mean statistics - Vegetation points (classes 3, 4, 5) height-normalized by subtracting ground elevation - Canopy defined as vegetation ≥ 2 m above ground (industry standard threshold) - Rasterized to 1 m grid cells; canopy cover = canopy cells / total boundary cells - Edmonton boundary from OpenStreetMap, reprojected to EPSG:3776

3.2 Individual Tree Delineation

Canopy Height Model (CHM) approach at 0.5 m resolution: 1. CHM generated as max vegetation elevation minus ground DEM 2. Gaussian smoothing ($\sigma = 1.5$ cells) to reduce noise 3. Local maxima detection for tree top identification (minimum spacing 2 m) 4. Watershed segmentation from tree tops to delineate individual crowns 5. Per-crown feature extraction: height, area, diameter, shape metrics

3.3 Species Classification

Conifer/Deciduous (Binary) Random Forest classifier (300 trees, $\text{max_depth}=20$) trained on 318,373 inventory-matched crowns with 45 LiDAR-derived features: - Height structure: percentiles, skewness, kurtosis, vertical distribution - RGB statistics from colorized LiDAR points - Intensity and return structure metrics - Crown shape: compactness, eccentricity, height-to-width ratio

Training labels derived automatically by spatial matching City tree inventory points (474K trees with genus) to LiDAR-delineated crowns within 5 m radius.

Top-5 Species LightGBM classifier on 239,040 training samples with 48 features (LiDAR + orthophoto RGB): - Classes: Ash, Aspen, Elm, Pine, Spruce - Automated Karpathy loop (1,600+ experiments) for hyperparameter optimization

3.4 FBP Fuel Type Classification

CFFDRS/FBP fuel types assigned within UPLVI natural and naturalized area boundaries: - Conifer fraction per 20 m cell computed from classified trees - Agricultural lands differentiated using UPLVI subtype classification - CanFG-style codes with embedded percent conifer for M-2 mixedwood

4. Results

4.1 Canopy Cover

Metric	Value
City land area	784.5 km ² (78,448 ha)
Total canopy area	119.6 km ² (11,956 ha)
Canopy cover	15.2%

Height Class Breakdown

Class	Cover	Area
Low canopy (2-5 m)	11.8%	9,290 ha
Medium canopy (5-15 m)	11.1%	8,705 ha
High canopy (>15 m)	1.9%	1,522 ha

Canopy Cover in Context

Source	Year	Canopy %
UFMP / UFORE model	2009	10.3%
This analysis	2025	15.2%
UFMP target	–	20%

4.2 Five-Class Vegetation Summary

Class	Count	Canopy Area (ha)
Tree Deciduous	2,826,067	13,984
Tree Coniferous	1,491,063	5,553
Shrub Deciduous	681,817	1,254
Shrub Coniferous	169,195	678

4.3 By Management Context

Context	Crowns	Canopy Area (ha)
Natural / Naturalized	2,158,048	7,565
Ornamental (planted)	3,010,094	13,904

4.4 Species Classification Performance

Species	F1 Score	Precision	Recall
Elm	0.839	0.852	0.827
Spruce	0.788	0.783	0.793
Ash	0.787	0.788	0.786
Aspen	0.664	0.651	0.677
Pine	0.618	0.604	0.632

4.5 FBP Fuel Type Distribution

Fuel Type	Code	Area (ha)	% of City
C-2 (Boreal Spruce)	2	506	0.6%
D-2 (Green Aspen)	12	2,129	2.7%
M-2 (Mixedwood Green)	14	1,746	2.2%
O-1a (Crops/Pasture)	31	16,984	21.7%
O-1b (Standing Grass)	32	7,644	9.7%
NF (Non-fuel)	99	49,434	63.0%

5. Deliverables

5.1 GIS Layers

GeoPackage: edmonton_canopy_analysis.gpkg (93 MB)

Layer	Type	Features
edmonton_boundary	Polygon	1
canopy_tiles	Polygon grid	3,228
tree_density_grid	Polygon grid	3,228
tree_crowns_500k	Points	500,000

5.2 BurnP3/Prometheus Package

File	Format	Description
fuel_type.asc + .lut + .prj	ASCII grid	FBP fuel codes (16 classes)
elevation.asc + .prj	ASCII grid	CDEM 30 m resampled to 20 m
fuel_type.tif	GeoTIFF	FBP fuel codes (for BurnP3+)
percent_conifer.tif	GeoTIFF	0-100% for M-2 cells

5.3 Data Files

File	Records	Description
canopy_tile_results.csv	3,228	Per-tile canopy statistics
individual_trees.csv	5,168,142	All delineated crowns
trees_classified_v2.csv	3,922,940	Crowns with species classification
crown_features.csv	3,922,940	42-feature profiles per crown

5.4 Models

Model	Accuracy	Description
classifier_v2.joblib	90.4%	Conifer/deciduous Random Forest
Top-5 species (LightGBM)	0.74 macro F1	Elm/Ash/Spruce/Aspen/Pine

6. Limitations and Recommendations

6.1 Limitations

1. **Grass classification** relies on UPLVI/naturalized boundaries rather than spectral analysis. Natural vs managed grass distinction is approximate.
2. **Species classification** accuracy varies by species. Pine (62%) and Aspen (66%) have lower F1 scores due to structural similarity with other species.
3. **Orthophoto coverage** is partial (37% of training data). Full coverage would improve species classification.
4. **Leaf-on only** – a single collection season. Leaf-off LiDAR would improve deciduous species separation and ground classification under canopy.

6.2 Recommendations

1. **Acquire 4-band CIR imagery** for future collections to enable NDVI-based vegetation classification.
 2. **Integrate neighbourhood features** for Elm detection to leverage spatial clustering patterns.
 3. **Repeat analysis with 2019 LiDAR** for temporal change detection (canopy gain/loss).
 4. **Ground-truth validation** of species classification with field crews in representative areas.
 5. **Update UFMP** canopy cover target tracking with the 15.2% baseline.
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7. Technical Specifications

Parameter	Value
CRS	EPSG:3776 (NAD83 / Alberta 3TM ref merid 114 W)
LiDAR point density	25 pts/m ²
Canopy raster resolution	1 m
Tree delineation CHM	0.5 m
Fuel raster cell size	20 m
Canopy height threshold	2.0 m
Tree/shrub boundary	5.0 m
Classifier training samples	318,373 (auto-matched from inventory)
